Tract-1

Q1) var: can be redeclared and reassigned

Let: can be reassigned, but can’t be redeclared

Const: can’t be redeclared or reassigned

Q2)

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Document</title>
    </head>
    <body>
        <script>
            let name = "Siya";
            let age = 20;
            if(age >= 18)
        {
            console.log("You are an adult");
        }
        else
        {
            console.log("You are a minor");
        }
            </script>
    </body>
</html>
```

```
You are an adult T1q2.html:14
```

If we use the string “18” instead, JS will do a type correction and if we change the condition to (age!==18) it will still print “You are an adult”.

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Document</title>
    </head>
    <body>
        <script>
            let name = "Siya";
            let age = "20";
            if(age!== 18)
        {
            console.log("You are an adult");
        }
        else
        {
            console.log("You are a minor");
        }
            </script>
    </body>
</html>
```

```
You are an adult
>
```

Q3)

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Document</title>
    </head>
    <body>
        <script>
            function buildScoreRegistry(){
                let scores = [77, 86, 91, 65, 80];
                scores.push(95);
                for(let i=0; i<scores.length; i++)
            {
                console.log(scores[i]);
            }
        }
            buildScoreRegistry();

            </script>
    </body>
</html>
```

```
77
86
91
65
80
95
```

Scores.shift method will remove the element from start

Since const does not allow reassignment, we cannot do scores =[ ] to make it empty

Q4)

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Document</title>
    </head>
    <body>
        <script>
            function createUserAccount(firstName, lastName, baselineScore){
                const user = {
                    firstName,
                    lastName,
                    fullName: `${firstName} ${lastName}`,
                    score: baselineScore,
                    upgradeScore(){
                        this.score += 10;
                    }
                };
                return user;
            }

            let userA = createUserAccount("Priti", "Sharma", 80);
            console.log("Full name= "+userA.fullName);
            console.log("Initial score: "+userA.score);
            userA.upgradeScore();
            console.log("Score now: "+userA.score);

             let userB = createUserAccount("Shreya", "Mehta", 75);
            console.log("Full name= "+userB.fullName);
            console.log("Initial score: "+userB.score);
            userB.upgradeScore();
            console.log("Score now: "+userB.score);

            </script>
    </body>
</html>
```

```
Full name= Priti Sharma
Initial score: 80
Score now: 90
Full name= Shreya Mehta
Initial score: 75
Score now: 85
>
```

No, user A does not affect user B

Q5)

Here, “this” refers to the handler object so it finds this.bonusModifier = 15

The exact number logged to the console = 50 + 15 = 65